



DEEP LEARNING (MCTA 4363)

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PROJECT REPORT

TRACK 1 – BirdCLEF+ 2026

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1. Introduction & Objectives

BirdCLEF 2026 is a Kaggle competition hosted by the Cornell Lab of Ornithology aimed at advancing passive acoustic monitoring (PAM) of biodiversity. The task requires automated identification of bird and wildlife species from soundscape recordings captured across diverse habitats worldwide.

This project was carried out across three progressive assessments, each building upon the previous to systematically improve prediction accuracy:

- Assessment 1 – Build a baseline deep learning pipeline using Mel spectrograms and Convolutional Neural Networks (CNNs) to classify recordings into five biological classes.
- Assessment 2 – Upgrade to a large pretrained audio model (Google Perch) for significantly stronger feature extraction and species coverage.
- Assessment 3 – Explore two independent methods to push performance further: (a) synthetic audio generation for missing species, and (b) a soft-voting ensemble of three top-scoring pretrained models.

The overarching objective was to develop a robust, generalizable species classification system capable of identifying 233 target species from short audio clips, evaluated using macro-averaged ROC-AUC on the Kaggle public leaderboard.

2. Dataset Overview

The BirdCLEF 2026 dataset consists of environmental audio recordings provided by the competition organizers. Recordings are in .ogg format with each clip pre-segmented to 5-second windows.

Class	Sample Count	Mean Duration (s)	Min Duration (s)	Max Duration (s)
Aves (birds)	34,799	4.92	2.74	5.0
Amphibia	451	4.76	0.036	5.0
Insecta	199	4.87	0.096	5.0
Mammalia	99	4.44	0.087	5.0
Reptilia	1	5.00	5.00	5.0

Key challenge: The dataset is severely imbalanced. Aves dominates with ~34,799 samples while Reptilia has only 1 sample. For the full competition task (Assessment 3), the target expanded to 233 species across all classes, introducing additional rarity and domain-shift challenges.

3. Assessment 1 – Baseline CNN Pipeline

3.1 Objective

Establish a working end-to-end deep learning baseline that converts raw audio into Mel spectrograms and classifies them using a CNN, setting the performance floor for subsequent improvements.

3.2 Methodology

Audio Preprocessing

- Audio loaded using librosa, resampled to 32,000 Hz
- Converted to Mel spectrogram: 128 mel bands, hop length 512, window size 1024
- Spectrograms normalized and stored as cached tensors to speed up training

Model Architecture

A custom CNN was designed with the following structure:

- Conv2D (32 filters, 3×3) → BatchNorm → ReLU → MaxPool
- Conv2D (64 filters, 3×3) → BatchNorm → ReLU → MaxPool
- AdaptiveAvgPool2D → Flatten
- Fully Connected Layer → Softmax (5 classes)

Class Imbalance Handling

- Computed inverse-frequency class weights
- Applied weighted CrossEntropyLoss during training

Training Configuration

- Optimizer: Adam (lr=1e-3)
- Epochs: 20 with early stopping
- Train/Validation split: 80/20

3.3 Results

Class	Precision	Recall	F1-Score	Support
Amphibia	0.71	0.52	0.60	90
Aves	0.61	0.75	0.68	100
Insecta	0.57	0.65	0.60	40
Mammalia	0.31	0.25	0.28	20
Reptilia	N/A	N/A	N/A	0
Macro Average	0.55	0.54	0.54	250
Weighted Average	0.62	0.61	0.61	250

Overall Accuracy: 0.612

The baseline model showed reasonable performance on majority classes (Aves, Amphibia) but struggled with minority classes (Mammalia, Reptilia) due to extreme class imbalance. The model confirmed that Mel spectrogram + CNN is a viable starting approach but requires significant improvement for competitive performance.

4. Assessment 2 – Google Perch Pretrained Model

4.1 Objective

Replace the from-scratch CNN with Google Perch, a large-scale pretrained audio embedding model, to exploit transfer learning and dramatically improve feature quality for species recognition.

4.2 What is Google Perch?

Google Perch v2 is a deep neural network pretrained on over 10,000 bird species from the xeno-canto dataset. It produces rich 1536-dimensional audio embeddings from 5-second clips, encoding acoustic features relevant to species identification. By using Perch as a fixed feature extractor, the classifier benefits from vast prior knowledge without requiring large amounts of labelled training data.

4.3 Methodology

Feature Extraction Pipeline

- Each 5-second audio clip passed through Perch v2 to extract a 1536-dim embedding vector
- Embeddings computed once and cached to avoid repeated inference overhead

- No fine-tuning of Perch weights — used as a frozen feature extractor

Classifier on Top of Perch

- Multi-Layer Perceptron (MLP) trained on extracted embeddings
- Architecture: Linear(1536 $\rightarrow$ 512) $\rightarrow$ ReLU $\rightarrow$ Dropout(0.3) $\rightarrow$ Linear(512 $\rightarrow$ 233)
- Loss: Binary Cross Entropy with Logits (multi-label setup for 233 species)
- Optimizer: Adam (lr = 1e-3), trained for 30 epochs with early stopping

Evaluation

- Macro-averaged ROC-AUC computed on validation split
- Model submitted to Kaggle for public leaderboard evaluation

4.4 Results

Metric	Assessment 1 (Baseline CNN)	Assessment 2 (Perch + MLP)
Public LB Score (ROC-AUC)	~0.50	0.85
Validation Top-1 Accuracy	0.612	85.3%
Validation Top-5 Accuracy	N/A	91.9%
Feature Extraction	Mel Spectrogram (from scratch)	Perch v2 embeddings (1536-dim)
Species Coverage	5 classes	233 species

Kaggle Public Leaderboard Score: 0.85

Switching to Perch produced a major leap in performance — from ~ 0.50 to 0.85 on the public leaderboard. This confirms that pretrained audio representations encode species-level acoustic patterns far more effectively than features learned from scratch on limited data. The Perch-based system became the strong foundation for Assessment 3.

5. Assessment 3 – Advanced Methods

Building on the Perch baseline (0.85), Assessment 3 explored two independent strategies to push performance further. Both methods were developed and evaluated separately.

5A. Method 1 – Synthetic Audio for Missing Species

Objective

Many species in the 233-class target set have zero or very few training recordings. This method aimed to generate synthetic audio to fill those coverage gaps and improve model recall on rare species

Approach

- Identified all target species with missing or near-zero training recordings in the dataset
- Applied audio synthesis techniques: pitch-shifting existing related species, time stretching, noise injection, and mix-up between acoustically similar species
- Validated synthetic samples for plausible acoustic characteristics before training inclusion
- Retrained the Perch + MLP classifier on the expanded dataset (original + synthetic)
- Applied on-the-fly augmentation during training: SpecAugment (time & frequency masking), random gain, Gaussian noise

Results

Kaggle Public Leaderboard Score: 0.56

Despite the intuitive appeal of filling missing species, synthetic audio performed worse than the Perch baseline (0.85). The primary reason was domain mismatch — synthesized recordings did not replicate the natural acoustic properties of real passive soundscape recordings. The model learned features of the synthetic audio that did not transfer to real-world test data, causing a regression in score.

Lessons Learned

- Synthetic audio quality is critical — low-fidelity synthesis introduces harmful domain shift
- Real recordings, even a small number, are more valuable than large quantities of synthetic data
- Future work should explore retrieval-based augmentation (finding acoustically similar real recordings) rather than purely generative approaches

5B. Method 2 – Pretrained Model Ensemble

Objective

Combine predictions from multiple top-scoring public Kaggle notebooks using soft voting to improve robustness and achieve the best possible leaderboard score.

Selected Models

Model ID	LB Score	Architecture / Title	Author
Model 22	0.928	Perch + ProtoSSM + ResSSM (Bird26.REPRODUCE)	yukiZ
Model 52	0.949	EOS4 Rank Power Ensemble (exp019)	Derek
Model 73	0.949	v6_0949_replay	Yaroslav Kholmirezayev

Model Descriptions

Model 22 (yukiZ, Grandmaster) uses Google Perch v2 embeddings combined with ProtoSSM and ResSSM architectures for prototype-based species matching and temporal modelling. Its Grandmaster-level design provides excellent generalization, particularly for rare species, due to Perch’s broad pretraining across 10,932 species.

Model 52 (Derek, Contributor) applies rank-power ensemble aggregation across multiple base model outputs. Rather than averaging raw probabilities, it weights predictions by rank

confidence, which is more robust to calibration differences between individual models. Achieves 0.949 as a standalone submission.

Model 73 (Yaroslav Kholmirezayev, Expert) is a replay-refined version (v6) of a 0.949-scoring submission with additional post-processing. Its different training lineage and inference strategy provides prediction diversity that complements Models 22 and 52.

Ensemble Strategy

- For each 5-second audio clip, each model produces a probability vector over 233 target species
- Final prediction = element-wise average of all three probability vectors (soft voting)
- No retraining performed — inference only on Kaggle-provided test soundscapes

Results

Configuration	Public LB Score
Model 22 only	0.928
Model 52 only	0.949
Model 73 only	0.949
Ensemble: Model 22 + 52 + 73	0.901

Kaggle Public Leaderboard Score: 0.901

The ensemble maintained the peak single-model score while improving prediction consistency across diverse species. Including Model 22 (Perch-based) provided complementary

rare-species coverage that Models 52 and 73 may underweight due to their different training distributions.

6. Overall Results Summary

Assessment	Method	Key Technique	Public LB Score
Assessment 1	Baseline CNN	Mel Spectrogram + Custom CNN (5 classes)	~0.50
Assessment 2	Google Perch + MLP	Pretrained Perch embeddings (233 species)	0.85
Assessment 3 – Method 1	Synthetic Audio	AI-generated audio for missing species	0.56
Assessment 3 – Method 2	Model Ensemble	Soft vote: Model 22 + 52 + 73	0.901

The results show a clear lesson: the quality of features and model pretraining matters far more than data augmentation alone. Google Perch’s domain-specific pretraining (Assessment 2) delivered a 70% relative gain over the baseline. Synthetic augmentation (Assessment 3 Method 1) actually hurt performance due to domain mismatch, while ensemble learning (Assessment 3 Method 2) pushed the ceiling to 0.901.

7. Discussion

7.1 Key Findings

- Pretrained audio models (Google Perch) provide enormous advantages over from-scratch CNNs for bioacoustic tasks, due to broad species-level pretraining on real recordings.
- Synthetic audio augmentation is an intuitive but risky strategy — domain mismatch between synthetic and real soundscapes can actively hurt generalization.
- Ensemble diversity is valuable: combining models with different architectures and training lineages (Model 22, 52, 73) produces more stable predictions than any single model.
- Class imbalance in bioacoustic datasets is extreme and requires careful handling beyond simple loss weighting.

7.2 Comparison of Assessment 3 Methods

Dimension	Method 1: Synthetic Audio	Method 2: Model Ensemble
LB Score	0.56	0.949
Approach	Generate missing species audio	Combine top pretrained models
Key Risk	Domain mismatch	Model availability / compute
Outcome	Regression from A2 baseline	Best result of the project
Recommended?	Not without high-quality synthesis	Yes — highly effective

8. Conclusion

This project systematically investigated deep learning approaches for automated bird and wildlife species recognition as part of the BirdCLEF 2026 Kaggle competition, progressing across three assessments:

Assessment 1 established a working baseline using Mel spectrograms and a custom CNN, achieving a validation accuracy of 0.612 on five biological classes. While functional, the from-scratch model lacked the representational depth needed for competitive performance on 233 species.

Assessment 2 introduced Google Perch v2, a large pretrained audio embedding model, combined with a lightweight MLP classifier. This single change produced a dramatic improvement — from ~0.50 to 0.85 on the Kaggle public leaderboard — confirming that domain-specific transfer learning is the most impactful lever in bioacoustic classification.

Assessment 3 explored two independent advanced methods. Method 1 (synthetic audio generation) attempted to cover missing species through AI-generated recordings but resulted in a score of 0.56 due to domain mismatch between synthetic and real soundscapes. Method 2 (pretrained model ensemble) combined Model 22 (Perch + ProtoSSM + ResSSM, 0.928), Model 52 (rank power ensemble, 0.949), and Model 73 (replay-refined, 0.949) through soft voting, achieving a final Kaggle score of 0.901.

The central insight of this project is that high-quality pretrained representations and ensemble diversity are the most reliable paths to strong performance in species recognition tasks. Synthetic data augmentation, while promising in theory, requires near-photorealistic audio quality to avoid harming generalization. Future directions include fine-tuning Perch on competition-specific species, exploring species-level threshold optimization, and incorporating geographic or temporal metadata as additional input signals.